

$$M_S M_e = 2.7 \times 10^7$$

10.

Ans: C

Since satellites are orbiting around the sun, the energies of the satellites can be represented by $E_p = -GM_S M/r$, $E_k = GM_S M/2r$ and $E_{total} = -GM_S M/2r$.

Since satellite A has a larger mass than satellite B and both satellites have the same kinetic energy,

$$GM_S M_A / 2r_A = GM_S M_B / 2r_B$$

$$M_A / r_A = M_B / r_B$$

$\rightarrow r_A$ is larger than r_B . (Option B is true)

By Kepler's 3rd law, since T^2 is proportional to r^3 , T_A is larger than T_B . (Option A is true)

Since $\omega = 2\pi/T$, ω_A is smaller than ω_B . (Option D is true)

Using $E_{total} = -GM_S M/2r$ and $M_A/r_A = M_B/r_B$,

both satellites should have the same total energy (Option C is false)